



HfO₂ FeRAM Production System

Master Report

University of Pristina × Infineon Collaboration

Ontologies Development Workstream

Project GitHub: <https://github.com/OPXHS/Ontologies-Development>

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Abstract

Process Ownership

Student	Assigned chapter	Position in the manufacturing flow
Olti Krasniqi	CMOS steps	Front end of line: access transistor, peripheral CMOS, and preparation for the high- k module
Art Haxholli	ALD HfO ₂ -based capacitor integration	High- k thin-film module within FEOL or memory-cell formation
Fatos Rama	Backend / BEOL integration	Contacts, multilevel interconnect, passivation, pads, and wafer-test handoff

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Chapter 1

Project Context and Research Question

Chapter 2

CMOS Steps

2.1 Module Purpose

2.2 Input Condition

2.3 Step-by-Step Flow

2.3.1 Starting Wafer and Cleaning

2.3.2 Isolation Definition

2.3.3 Well and Channel Formation

2.3.4 Gate Dielectric and Gate Electrode

2.3.5 Gate Patterning and Etch

2.3.6 Source/Drain Engineering

2.3.7 Spacers and Dopant Activation

2.3.8 Silicide and Contact Formation

2.3.9 Repeated Patterning Cycle

Table 2.1: Ordered CMOS process steps

Order	Process step	Purpose	Method	Equipment type	Output /next input
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2.4 Key Materials and Equipment

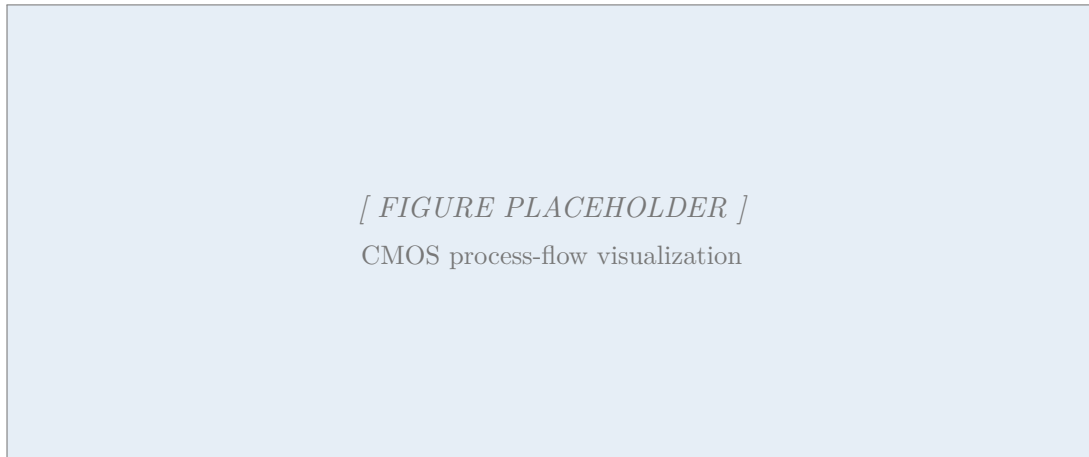


Figure 2.1: CMOS module process flow

Table 2.2: CMOS materials and equipment

Step	Material or chemical	Equipment or capability	Role in the module
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2.5 Critical Parameters

Table 2.3: CMOS critical parameters

Step	Parameter	Unit	Value or range	Evidence status /source
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2.6 Metrology and Electrical Tests

2.6.1 Critical-Dimension Measurement

2.6.2 Overlay Metrology

2.6.3 Film-Thickness Measurement

2.6.4 Sheet-Resistance Measurement

2.6.5 Electrical Parametric Tests

Table 2.4: CMOS metrology and electrical tests

Step or checkpoint	Measurement or test	Acceptance measure	Decision or process response
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2.7 Ontology Model and Communication

2.7.1 Protégé Ontology Design

2.7.2 Classes and Properties

Table 2.5: CMOS ontology classes and properties

Ontology element	Element type	Domain /parent	Range /definition
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2.7.3 Individuals and Example Records

Table 2.6: CMOS ontology individuals

Individual	Class	Key assertions	Source record
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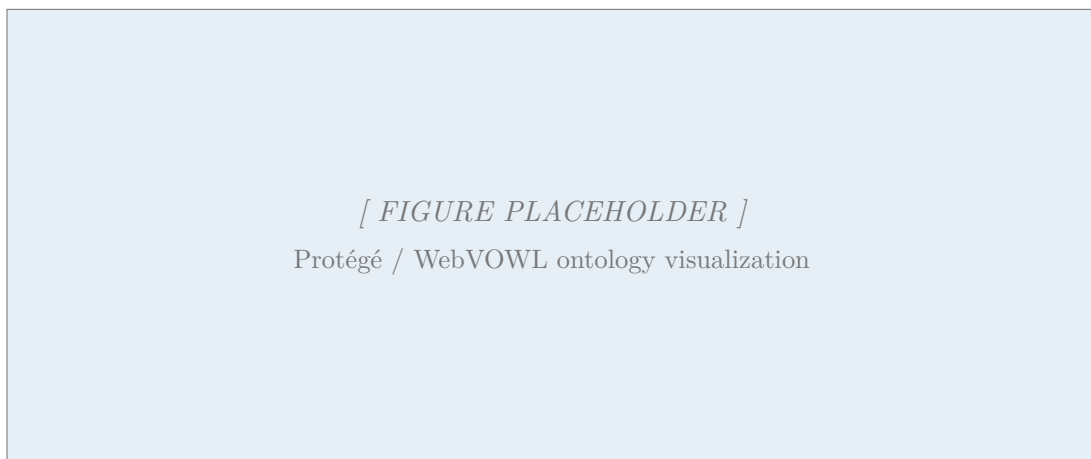


Figure 2.2: CMOS ontology visualization

2.7.4 Digital Reference Mapping

Table 2.7: CMOS Digital Reference mappings

Local concept	Digital Reference IRI	Mapping relation	Evidence and rationale
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2.7.5 Reasoner and Validation Results

2.7.6 Ontology Limitations

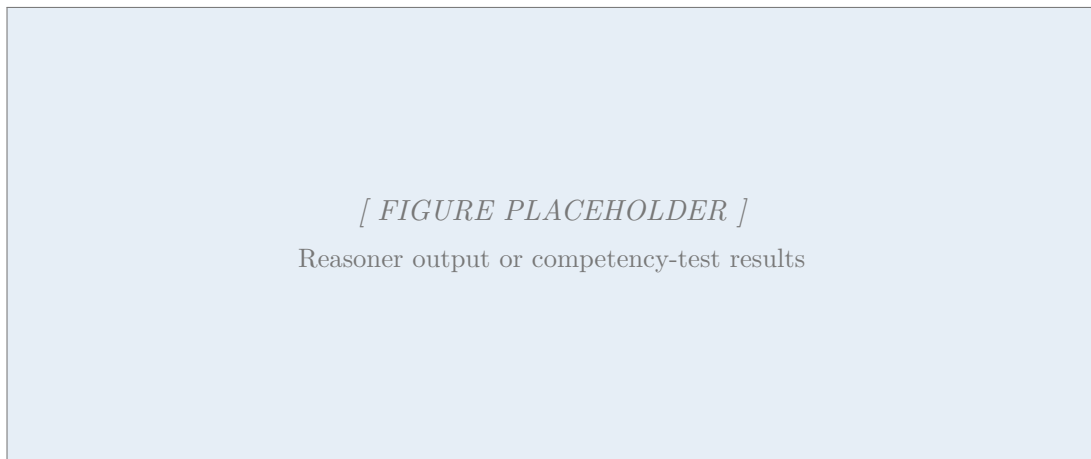


Figure 2.3: CMOS ontology validation results

Chapter 3

ALD HfO₂ Deposition

3.1 Starting Conditions and Predeposition Preparation

3.1.1 Starting Film, Substrate, and Device Conditions

Starting conditions are bounded by the reported HZO composition, deposited thickness, wafer temperature, and capacitor diameter, so comparisons remain configuration-specific rather than sharing a universal baseline [kim2020tin, kim2014inductive]. One study defines a single equimolar Hf:Zr deposition condition and device-size range, while another contributes only a thicker-film condition [kim2020tin, kim2014inductive]. The five-row evidence base spans just these two sources; it does not support broader claims about substrate material, device architecture, or alternative starting films [kim2020tin, kim2014inductive].

3.1.2 Surface Cleaning and Exposure Control

Reported surface preparation is not a single interchangeable clean: one substrate flow uses SPM followed by RCA SC-1/SC-2, while another uses sequential ultrasonic acetone, ethanol, and DI-water steps [park2022chamber, park2023seed]. Vacuum-break exposure is separately associated with increased carbon and tetragonal-phase stabilization, making exposure history part of the starting condition [schroeder2023road]. Other documented wet-chemical steps occur after electrode formation and therefore cannot define a common predeposition protocol [choi2025purge, kim2020tin, neuber2022peald].

3.1.3 Interface and Seed-Layer Engineering

Interface preparation is an explicit design variable: reported interlayers are introduced to alter crystallization, limit interlayer formation or defect migration, interrupt HZO grain-boundary paths, and buffer oxygen exchange [schroeder2023road, liao2023postmoore]. Al₂O₃ can serve as an internal dielectric or diffusion barrier, but the reported tradeoff includes voltage division and an added depolarization field; Pt insertion instead targets metal/oxide interfacial-layer suppression and a higher interface barrier [schroeder2023road, liao2023postmoore, kashir2021wake]. The seed-layer study retains top, bottom, dual, and unseeded configurations as distinct specimens rather than pooling them, with the dual ZrO₂ layers directly resolved near the interfaces [park2023seed]. Likewise, HfO₂/ZrO₂ superlattices remain sensitive to the starting layer and sublayer thickness, while a Zr-rich middle layer defines a separate trilayer architecture [schroeder2023road, liu2025zrrich]. For Si-doped stacks, the slightly thicker PEALD SiO₂ monolayer relative to thermal ALD shifts the doping concentration, so nominal cycle ratios do not guarantee equivalent composition [neuber2022peald].

3.2 ALD Film-Formation Sequence

3.2.1 Hf-Precursor Delivery and First Half-Cycle

Precursor delivery is represented as a set of process-specific first half-cycles, not as one transferable Hf dose [song2021ferro, choi2025purge, kaatranen2026unif]. The supported routes span cyclopentadienyl, alkylamide, TEMA-, and TDMA-based precursors; the alkylamide routes are reported for thermal and plasma-enhanced ALD with limited thermal stability, while organometallic Hf precursors bearing organic ligands are associated with residual impurities [song2021ferro, schroeder2023road]. Process-specific delivery detail is concentrated in the TEMA-based PEALD study, where source and manifold conditioning, Ar-assisted dosing at fixed chamber pressure, and unequal Hf and Zr pulse lengths form one coupled recipe [choi2025purge]. Those delivery values therefore remain attached to that reactor and chemistry rather than serving as defaults for the other precursor families [choi2025purge]. The remaining temperature observations are likewise conditional: thermal and plasma-enhanced studies pair their substrate settings with particular reactors, films, and precursor systems [choi2025purge, park2023seed, kim2014inductive, neuber2022peald, kaatranen2026unif, liu2025zrrich, kashir2021wake]. Accordingly, precursor identity, delivery conditioning, reactor mode, and the surrounding film specification are treated as one first-half-cycle definition [song2021ferro, choi2025purge].

3.2.2 Purge and Co-Reactant Half-Cycle

After the metal dose, first-purge evidence is confined to one HZO PEALD study [choi2025purge]. That flow uses Ar at the same reported chamber pressure as precursor dosing and varies the precursor-purge duration as an experimental series; it does not establish a purge recipe for the other reactor or precursor variants [choi2025purge]. Co-reactant coverage is broader but remains route-specific [song2021ferro, schroeder2023road]. Water is reported in thermal ALD flows, ozone appears at study-specific concentrations and pulse lengths, and plasma-enhanced routes use oxygen-based exposures [kim2020tin, kim2014inductive, neuber2022peald, kaatranen2026unif, choi2025purge, park2023seed, liu2025zrrich, kashir2021wake]. In one PEALD sequence, the Hf and Zr oxygen-plasma steps receive different power–duration pairs, whereas the silicon-doping PEALD flow couples oxygen with a capacitively coupled Ar plasma [choi2025purge, neuber2022peald]. The oxidant step therefore remains paired with the relevant metal or dopant subcycle [choi2025purge, neuber2022peald]. Second-purge coverage is limited to one entry identifying nitrogen as purge and carrier gas for thermal HZO ALD; no supported second-purge duration, pressure, or broader mechanism is available [kim2014inductive].

3.2.3 Cycle Architecture and Repetition

Cycle repetition is configured around the intended composition and thickness rather than one fixed template [park2022chamber, choi2025purge, park2023seed, kaatranen2026unif]. In one thermal HZO series, the supercycle count changes with chamber temperature while the deposited thickness remains 10 nm, so the count is not temperature-independent [park2022chamber]. A PEALD route instead uses a defined Hf:Zr dose ratio and supercycle total to approach the same thickness [choi2025purge]. Another nanolaminate series holds the combined HfO₂ and ZrO₂ count at 120 cycles while changing the block length, separating sequence architecture from total cycle count [park2023seed]. The SS/SL comparison similarly reaches a common thickness target through alternating single cycles or grouped symmetric blocks [kaatranen2026unif]. Discrete-feeding ALD adds another conditional architecture by repeating the precursor half-cycle before the reactant half-cycle [schroeder2023road]. Repetition count, block size, half-cycle

duplication, and deposition condition therefore remain jointly specified [**schroeder2023road**, **park2022chamber**, **park2023seed**].

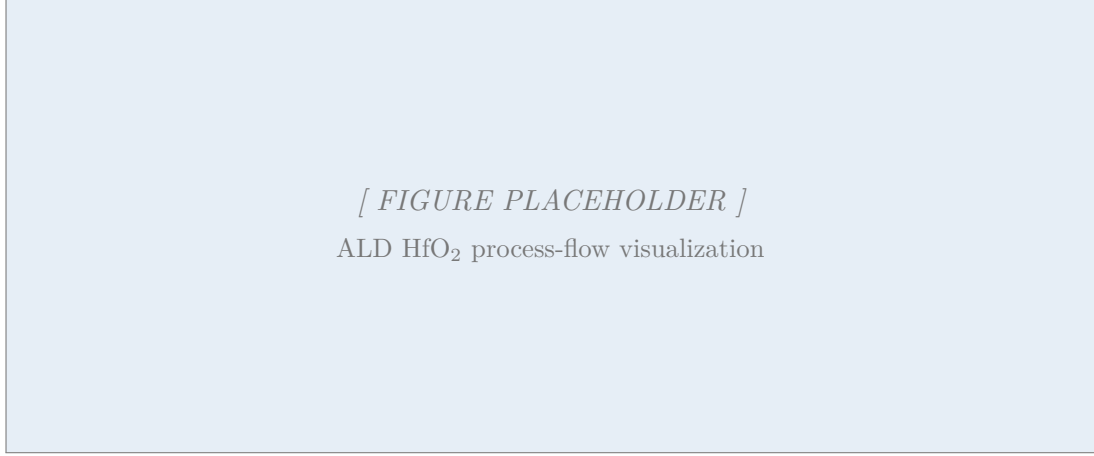
Table 3.1: Documented ALD cycle chemistries and operating windows

Film/stack	Hf precursor	Co-reactant	Cycle/timing	T/P window	Source/condition
HfO ₂	Cp ₂ HfMe ₂ ; (MeCp) ₂ HfMe ₂ ; (MeCp) ₂ Hf(OMe)Me; Hf(NEtMe) ₄ ; Hf(NMe ₂) ₄	H ₂ O, O ₂ plasma, O ₃	Not covered in sources	250–300 °C stability	Precursor and oxidant survey [song2021ferro]
HfO ₂ /ZrO ₂	TEMAHf	H ₂ O, O ₃ , O ₂ plasma, H ₂ O ₂	DF-ALD repeats precursor half-cycle	Not covered in sources	DF-ALD review [schroeder2023road]
TiN/10 nm Hf _{0.5} Zr _{0.5} O ₂ / TiN	Not covered in sources	H ₂ O	56–68 supercycles	120–250 °C	Thermal ALD chamber study [park2022chamber]
10 nm Hf _{0.5} Zr _{0.5} O ₂	TEMA-Hf	O ₂ plasma, 3–6 s	1.0/1.5 s doses; 3–90 s Ar purge	260 °C; 80 mTorr	3:2, 17-supercycle PEALD study [choi2025purge]
Hf _{0.5} Zr _{0.5} O ₂	TDMA-Hf	O ₃ , 400 g m ⁻³	Not covered in sources	Not covered in sources	High-concentration O ₃ ALD [kim2020tin]
Hf _{0.5} Zr _{0.5} O ₂ nanolaminate	TEMA-Hf	O ₂ plasma, 200 W	(5, 5) × 12–(60, 60) × 1; 120 cycles	250 °C	Seed-layer PEALD study [park2023seed]
Hf _{0.5} Zr _{0.5} O ₂	Not covered in sources	H ₂ O	N ₂ second purge	200 °C	Thermal ALD study [kim2014inductive]
10 nm HfO ₂ / SiO ₂	HfCl ₄	H ₂ O	10:1–27:1 HfO ₂ :SiO ₂ cycles	300 °C	Thermal HSO process [neuber2022peald]
Hybrid HfO ₂ / SiO ₂	Not covered in sources	O ₂ , Ar plasma	1 SiO ₂ :16–32 HfO ₂ cycles	50 °C SiO ₂ PEALD	Hybrid HSO process [neuber2022peald]
5.5 nm SS-/SL- Hf _{0.5} Zr _{0.5} O ₂	TDMAHf	H ₂ O	SS (1 + 1) × 27; SL 5(5 + 6) + 5	200 °C	Thermal super-lattice ALD [kaatranen2026unif]
Hf _{0.5} Zr _{0.5} O ₂ / ZrO ₂ / Hf _{0.5} Zr _{0.5} O ₂	TEMAH	Oxygen plasma	Not covered in sources	250 °C	Zr-rich stack PEALD [liu2025zrrich]
10 nm Hf _{0.5} Zr _{0.5} O ₂ on W	Not covered in sources	O ₃ , 276 g Nm ⁻³	1:1 Hf:Zr; 2–15 s O ₃ pulse	250 °C; 90 °C precursors	O ₃ -pulse ALD study [kashir2021wake]

3.3 Post-Deposition Processing

3.3.1 Post-Deposition Thermal Treatment

Post-deposition treatment is defined by the state of the stack at heating, not by an anneal label alone. PDA denotes uncapped RTA, whereas PMA follows TiN capping; elsewhere, PDA precedes upper-electrode deposition and PMA follows the top electrode [**song2021ferro**, **park2022chamber**, **park2023seed**]. The anneal sequence and electrode boundary condition are therefore coupled [**song2021ferro**, **park2023seed**]. A direct capped–uncapped comparison likewise places RTP after Ni formation for the capped MIM structure and before top-electrode deposition for the uncapped sample, so the two cases cannot share one recipe description [**kim2014inductive**].

Figure 3.1: ALD HfO₂ module process flow

Reported sequences impose still narrower placement constraints. One flow anneals after TiN top-electrode patterning but before bottom-electrode localization to minimize interface damage, whereas another performs RTA after bottom-TiN deposition and before HZO deposition specifically to isolate stress from the top electrode [kaatranen2026unif, kim2020tin]. Nitrogen RTA series, hydrogen-containing furnace treatment, high-pressure oxygen annealing, and NH₃ plasma pretreatment belong to different configurations or stated objectives; ambient therefore cannot be detached from sequence and purpose [park2022chamber, kim2020tin, liao2023postmoore]. The BEOL-constrained route additionally requires the entire process, including thermal treatment, to remain below 400 °C, making higher-temperature crystallization flows inapplicable to that stated constraint [schroeder2023road, neuber2022peald].

Table 3.2: Reported post-deposition treatment conditions

Film/stack	Anneal sequence	Temperature	Time	Ambient	Source/condition
HfO ₂	PDA; uncapped RTA	Not covered in sources	Not covered in sources	Not covered in sources	PDA definition [song2021ferro]
HfO ₂	PMA after TiN capping	Not covered in sources	Not covered in sources	Not covered in sources	PMA definition [song2021ferro]
HfO ₂ -ZrO ₂	Low-temperature crystallization	< 400 °C	Not covered in sources	Not covered in sources	BEOL compatibility [schroeder2023road]
Hf _{0.5} Zr _{0.5} O ₂	PMA by RTA	400–700 °C	30 s	N ₂	Temperature series [park2022chamber]
Hf _{0.5} Zr _{0.5} O ₂	RTA; 50 °C/s ramp	600 °C	30 s	N ₂ , 1 atm	PEALD capacitor stack [choi2025purge]
TiN/ Hf _{0.5} Zr _{0.5} O ₂ / TiN	PMA by RTA	400 °C	60 s	N ₂	Top-electrode anneal [kim2020tin]
TiN/ Hf _{0.5} Zr _{0.5} O ₂ / TiN	Forming-gas furnace anneal	400 °C	30 min	5% H ₂ + 95% N ₂	Hydrogen treatment [kim2020tin]
Hf _{0.5} Zr _{0.5} O ₂	PDA before upper electrode	450–650 °C	30 s	N ₂	Seed-layer study [park2023seed]
Ni-capped Hf _{0.5} Zr _{0.5} O ₂ MIM	RTP after top-electrode deposition	823 K	30 s	N ₂	Capped structure [kim2014inductive]
Uncapped Hf _{0.5} Zr _{0.5} O ₂ MIM	RTP before top-electrode deposition	Not covered in sources	Not covered in sources	Not covered in sources	Uncapped comparison [kim2014inductive]

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Table 3.2: Reported post-deposition treatment conditions (*continued*)

Film/stack	Anneal sequence	Temperature	Time	Ambient	Source/condition
Si:HfO ₂	Anneal after stack formation	800 °C	20 s	N ₂	Polar-phase formation [neuber2022peald]
TiN/HfO ₂ /ZrO ₂ /TiN	After TE patterning	470 °C	30 s	N ₂	Wafer-scale capacitors [kaatranen2026unif]
Conventional and Zr-rich Hf _{0.5} Zr _{0.5} O ₂	Rapid thermal annealing	400–450 °C	60 s	N ₂	Stack comparison [liu2025zrrich]
W/Hf _{0.5} Zr _{0.5} O ₂ /W	Rapid annealing	500–800 °C	1–30 s	N ₂	15 s O ₃ dosage [kashir2021wake]
HfO ₂ capacitor	High-pressure oxygen anneal	Not covered in sources	Not covered in sources	High-pressure O ₂	Wake-up suppression [liao2023postmoore]
HfO ₂ capacitor	NH ₃ plasma pre-treatment	Not covered in sources	Not covered in sources	NH ₃ plasma	Endurance strategy [liao2023postmoore]
Hf _{0.5} Zr _{0.5} O ₂ /TiN	Sequential O ₂ –H ₂ plasma oxidation	Not covered in sources	Not covered in sources	O ₂ –H ₂ plasma	TiO _x N _y growth [schroeder2023road]

3.3.2 Electrode and Stack Integration

Electrode integration is an ordered stack-construction problem because a nominal conductor does not specify its position, deposition history, or interface exposure [schroeder2023road, neuber2022peald]. The SNAP TiN/HZO/TiN route forms the electrodes and oxide in one ALD tool without a vacuum break, while another TiN stack uses ALD for the bottom electrode and PVD for the top electrode; these routes impose different interface histories [schroeder2023road, neuber2022peald].

Position-specific processing also changes downstream boundary conditions [schroeder2023road, kim2020tin]. For TiN, bottom-electrode oxidation promotes the ferroelectric phase through oxygen scavenging, while top-electrode microtexture depends on deposition pressure; a room-temperature top layer suppresses monoclinic-phase formation through stress-induced crystallization, whereas high-temperature top-electrode deposition is a reported limitation [schroeder2023road, kim2020tin]. Top and bottom TiN designations are therefore not interchangeable [kim2020tin]. In the W-based purge study, the bottom electrode, top capping layer, and Pt top contact are separate operations, with the Pt contact introduced for electrical characterization rather than serving as the W cap [choi2025purge]. Likewise, Au/Pd functions as a hard mask, and Ti/Au forms contacts to TiN electrodes; neither description defines the capacitor electrode stack by itself [kim2020tin, kaatranen2026unif]. Material substitutions retain their own conditions: RuO₂ and IrO₂ supply oxygen and improve endurance relative to TaN and TiN, whereas W is tied to its thermal-expansion-driven tensile-strain behavior [schroeder2023road]. The hybrid W/Pt/HZO/W configuration explicitly assigns top W and bottom Pt to combine their advantages, so collapsing it to a generic W/Pt designation loses the reported stack orientation [kashir2021wake].

Table 3.3: Electrode integration routes and stack conditions

Stack position	Electrode	Integration method	Thickness	Process condition	Source/configuration
Three-dimensional capacitor	Not covered in sources	Conformal ALD	10 nm	High-aspect 1T–1C FeRAM	Song configuration [song2021ferro]

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Table 3.3: Electrode integration routes and stack conditions (*continued*)

Stack position	Electrode	Integration method	Thickness	Process condition	Source/configuration
Bottom and top	TiN	In-tool SNAP ALD	Not covered in sources	Without vacuum break	Roadmap configuration [schroeder2023road]
Bottom and top	TiN	DC sputtering	50 nm	Reference capacitor	Choi configuration [park2022chamber]
Bottom	W	DC magnetron sputtering	Not covered in sources	3.3 W/cm ² ; 5.0 mTorr Ar	W baseline [choi2025purge]
Top capping	W	DC magnetron sputtering	20 nm	Reduced deposition time	W capping configuration [choi2025purge]
Top contact	Pt	Shadow-mask deposition	50 nm	Electrical characterization	Pt contact configuration [choi2025purge]
Bottom and top	TiN	RF sputtering	90 nm	250 W; Ar:N ₂ 20:1; RT or 500°C	TiN temperature study [kim2020tin]
Hard mask	Au/Pd	Sequential e-beam evaporation	85/3 nm	Sequential deposition	Hard-mask configuration [kim2020tin]
Upper	TiN	RF sputtering	50 nm	400 μ m; 29 ohm/sq	Seed-layer device [park2023seed]
Bottom stack	TiN/Ru	Not covered in sources	20/100 nm	300 nm SiO ₂ base	Induced-crystallization capacitor [kim2014inductive]
Top	Ni	Sputtering	50 nm	50 W cluster system	Ni-top capacitor [kim2014inductive]
Bottom and top	TiN	Bottom ALD; top PVD	10 nm	Silicon-doped hafnia capacitor	Neuber configuration [neuber2022peald]
Bottom and top	TiN	Reactive sputtering	30 nm	Superlattice capacitor	Wafer-scale configuration [kaatranen2026unif]
Electrical contacts	Ti/Au	Evaporation and lift-off	Not covered in sources	Contacts to TiN electrodes	Wafer-scale configuration [kaatranen2026unif]
Bottom and top	W	Magnetron sputtering	30 nm	Room temperature; top 80×80 μ m ²	Zirconium-rich capacitor [liu2025zrrich]
Bottom and top	W	Bottom sputter; top RF sputter	50 nm	Room temperature; 30×30–100×100 μ m ²	Wake-up-free capacitor [kashir2021wake]
Hybrid interfaces	Top W; bottom Pt	RTA-integrated stack	Not covered in sources	Mechanical stress with Pt interface	Hybrid capacitor [kashir2021wake]
Gate stack	Polysilicon/TiN	Not covered in sources	Not covered in sources	28 nm CMOS FeFET stack	FeFET configuration [mikolajick2018mrs]

3.4 Materials, Reactor Platforms, and Equipment

Reactor coverage is heterogeneous rather than interchangeable: the ledger records thermal ALD and PEALD, a single-wafer reactor, and a laminar-flow side-inlet reactor under different HZO growth conditions [park2022chamber, park2023seed, kim2014inductive, kaatranen2026unif, liu2025zrrich]. Equipment identity is unevenly resolved: Oxford FlexAL II, iOV dx2, and TFS 200 Beneq are named for specified deposition cases, whereas the thermal-ALD, single-wafer, and laminar-flow reports provide no manufacturer or model; Savannah

S100 is named without a reported condition [choi2025purge, kim2020tin, park2023seed, kim2014inductive, kaatranen2026unif, liu2025zrrich].

Comparison is bounded by reported specimen formats and measurement chains. A 6400 μm^2 patterned electrode area and a 150 mm wafer carrying 25–100 μm -diameter capacitors remain distinct descriptors [park2022chamber, kaatranen2026unif]. Spectroscopic ellipsometry and XPS are tied respectively to thickness control and unannealed-sample stoichiometry, while electrical measurements use a paired Radiant Workstation and Agilent B1500 [kim2014inductive, liu2025zrrich].

Table 3.4: Materials, reactor platforms, and measurement equipment

Equipment class	Tool/reactor	Application	Wafer/film format	Source/condition
Deposition reactor	Thermal ALD reactor	HZO deposition	Not covered in sources	HZO deposition process [park2022chamber]
Deposition reactor	Oxford FlexAL II	PE-ALD HZO deposition	Not covered in sources	PE-ALD HZO film [choi2025purge]
Deposition reactor	Savannah S100 ALD	Not covered in sources	Not covered in sources	Condition unspecified [kim2020tin]
Deposition reactor	iOV dx2 PEALD	HZO thin-film growth	50 nm TiN on oxide/silicon	iSAC growth; TiN sputtering [park2023seed]
Deposition reactor	Single-wafer ALD reactor	HZO thin-film growth	Single-wafer processing	HZO growth methods [kim2014inductive]
Deposition reactor	Laminar-flow side-inlet ALD reactor	SS- and SL-HZO growth	6400 μm^2 electrodes; 150 mm wafers	Patterning; SS/SL comparison [park2022chamber, kaatranen2026unif]
Deposition reactor	TFS 200 Beneq PEALD	Stack growth	HZO/ZrO ₂ /HZO stack	250 C substrate [liu2025zrrich]
Annealing equipment	Allwin21 AccuThermo AW 610	Rapid thermal annealing	HZO film	Rapid thermal anneal [choi2025purge]
X-ray metrology	Rigaku SmartLab X-ray diffractometer	GIXRD and XRR	HZO film	Cu K-alpha radiation [choi2025purge]
Thickness/composition metrology	Spectroscopic ellipsometry; XPS	Thickness control; stoichiometry monitoring	ALD HZO; unannealed samples	Methods measurements [kim2014inductive]
Electrode deposition	PVD 75 sputtering system	Bottom and top W deposition	Not covered in sources	W-electrode sputtering [liu2025zrrich]
Electron microscopy	Helios G4 FIB; Talos F200x	Lamella preparation; HR-TEM/STEM	Not covered in sources	30–200 kV operation [liu2025zrrich]
Electrical characterization	Radiant Workstation; Agilent B1500	Ferroelectric and parameter measurements	Not covered in sources	Paired electrical instruments [liu2025zrrich]

3.5 Critical Process and Device Parameters

3.5.1 Film Thickness and Growth per Cycle

Cycle-count interpretation must retain chemistry and temperature: independent HfO₂/ZrO₂ calibrations use TEMA-Hf/TEMA-Zr in PEALD or TEMA_H/TEMA_Z with O₃ at 250 °C, and HZO growth per supercycle decreases as chamber temperature rises from 120 to 250 °C [choi2025purge, park2022chamber, park2023seed, kashir2021wake]. Consequently, the ellipsometric thickness measured on Si reference pieces and the ALD-HZO thickness series

describe different specimens rather than a shared thickness window [kim2014inductive, kaatranen2026unif]. For Si-containing stacks, the standard thermal-ALD SiO₂ doping-layer process deposits significantly more slowly than PEALD, so the rate comparison remains specific to thermal-versus-plasma SiO₂ layer formation [neuber2022peald].

3.5.2 Conformality and Uniformity

Conformality evidence applies to a 10 nm ferroelectric-HfO₂ MFM capacitor in a high-aspect-ratio 3D 1T–1C geometry; the step-coverage claim is comparative to PVD/CVD, so it does not establish blanket wafer uniformity [song2021ferro]. For standard thermal and hybrid plasma-enhanced Si:HfO₂ processes, thickness differs across cycling ratios even though both show similar thickness standard deviations, while the absence of crossflow patterns is reported only for hybrid layers [neuber2022peald]. Wafer-scale interpretation depends on sampling 140 sites of $1 \times 1 \text{ cm}^2$ and separating center-area from full-wafer results at $\pm 3.0 \text{ V}$ [kaatranen2026unif]. A precursor-inlet polarization minimum coincides with thicker HZO, whereas the single failed SL site is attributed to incomplete Ti/Au removal rather than the film [kaatranen2026unif].

3.5.3 Polarization and Coercive Field

Polarization and coercive-field comparisons depend on whether the reported quantity is pristine, wake-up-conditioned, hysteresis-derived, or PUND-corrected [song2021ferro, kim2020tin, kashir2021wake]. In the TiN/HZO/TiN chamber study, pristine P_r changes across both deposition and anneal temperatures, and deposition at 120 °C gives significantly lower P_r than at 200 or 250 °C; anneal comparisons therefore require a matched chamber temperature [park2022chamber]. Thickness adds a separate constraint because ALD HZO shows increasing P_r and decreasing E_c as film thickness increases [kim2014inductive]. For $(20, 20) \times 3$ nanolaminates annealed at 550 °C, unseeded and top-seeded specimens form one similar P–E pair, whereas bottom-seeded and dual-seeded specimens form another; the 650 °C anneal degrades ferroelectric properties through monoclinic-phase formation, and greater nanolaminate thickness trades higher remanent polarization against poorer durability [park2023seed]. For W/HZO/W processed with a 15 s ozone pulse and a 700 °C/5 s anneal, pristine and woken values remain distinct [kashir2021wake]. Under the 700 °C/5 s anneal, PUND-corrected $2P_r$ differs between W/HZO/W and W/Pt/HZO/W [kashir2021wake]. In sub-6 nm HZO/Zr-RL/HZO at 1.0 V, the $2P_r$ /coercive-voltage pair belongs to the same operating condition [liu2025zrrich]. Separately, the 25–125 °C temperature series reports stable $2P_r$ without a significant coercive-voltage shift for that stack [liu2025zrrich].

3.5.4 Leakage-Current Constraints

Leakage comparisons require matched bias, conditioning, and stack [park2022chamber, kaatranen2026unif]. For TiN/HZO/TiN, leakage measured at -3 V rises with anneal temperature at fixed chamber temperature and with chamber temperature at fixed anneal temperature [park2022chamber]. The SS/SL comparison instead reports maxima after wake-up at $\pm 3.0 \text{ V}$, so those results do not define a common limit with the chamber/anneal series [kaatranen2026unif]. The current reported at 12 V for a $1.1 \times 10^{-3} \text{ cm}^2$ electrode is identified as domain-switching current, not leakage current [kim2014inductive].

3.5.5 Defect Density, Endurance, and Reliability

Reliability outcomes remain specific to process history, stack, and electrical stress rather than defining a transferable cycle-count window [park2022chamber, choi2025purge, kaatranen2026unif, liu2025zrrich]. In purge-series HZO capacitors, a 3 s precursor purge shows no fatigue or breakdown through 10^9 cycles, whereas the 30 and 90 s variants break at

2×10^8 and 6×10^7 cycles, respectively [choi2025purge]. The chamber-temperature series presents a different tradeoff: all tested chamber/anneal combinations avoid breakdown through 10^8 cycles, yet fatigue begins earlier as chamber temperature rises; the accompanying oxygen-vacancy increase is linked to higher P_r and a lower critical cycle count [park2022chamber]. In SS/SL wafers, increasing cycling voltage shortens breakdown life for both architectures and SL outlasts SS at matched voltage; after 10^6 cycles, post-wake-up retention loss remains below 4% through 10^4 s [kaatranen2026unif]. At a saturated frequency of 100 kHz, endurance decreases by one decade for each 0.22 V increase [liu2025zrrich]. For the HZO/Zr-RL/HZO stack, endurance reaches up to 10^{11} cycles at cycling voltages no greater than 1.5 V; the approximately 10^{11} -cycle result at 1.0 V is explicitly breakdown-limited [liu2025zrrich]. The 10-year retention result for that stack belongs to 1.0 V and 125 °C [liu2025zrrich]. Reported FeFET, MFM-capacitor, and FeRAM-array endurance values remain configuration-specific because their stacks, fields, frequencies, and operating temperatures differ [liao2023postmoore].

Table 3.5: Critical process and device parameter windows

Parameter	Range/result	Film/stack	Process condition	Limitation	Source
HfO ₂ growth rate	1.00–1.18 Å/cycle	Independent HfO ₂ films	TEMA-Hf or TEMAH precursor processes	Not covered in sources	[choi2025purge, park2023seed, kashir2021wake]
ZrO ₂ growth rate	1.00–1.34 Å/cycle	Independent ZrO ₂ films	TEMA-Zr or TEMAZ precursor processes	Not covered in sources	[choi2025purge, park2023seed, kashir2021wake]
Film thickness	5.5–25 nm	Hf _{0.5} Zr _{0.5} O ₂ and SS/SL HZO	ALD; ellipsometry reference pieces	Not covered in sources	[kim2014inductive, kaatranen2026unif]
Si-doping-layer deposition rate	PEALD significantly exceeds thermal ALD	Hybrid versus standard HSO	PEALD versus thermal ALD	Not covered in sources	[neuber2022peald]
Conformal high-aspect coverage	13:1 aspect ratio	10 nm FE-HfO ₂ capacitor	3D 1T-1C FeRAM	Not covered in sources	[song2021ferro]
Step coverage	Excellent for high-aspect structures	ALD HfO ₂ films	ALD versus PVD/CVD	Polycrystalline films include nonferroelectric phases	[song2021ferro]
Process-dependent thickness	10.6–12.7 nm	Standard HSO; hybrid HSO	All tested cycling ratios	Not covered in sources	[neuber2022peald]
Thickness nonuniformity	3.2–3.7% standard deviation	Standard and hybrid HSO	Thermal ALD and hybrid PEALD	Not covered in sources	[neuber2022peald]
Effective sublayer thickness	0.5 nm	SL-HZO: ZrO ₂ /HfO ₂	5 ZrO ₂ plus 6 HfO ₂ cycles	Not covered in sources	[kaatranen2026unif]
Wafer-scale uniformity improvement	18–53%	SS/SL HZO, 150 mm wafers	70×70 mm ² center; ±3.0 V	One SL site failed during lift-off	[kaatranen2026unif]
Maximum leakage-current density	0.04–0.08 A/cm ²	SS/SL HZO capacitors	±3.0 V after wake-up	Not covered in sources	[kaatranen2026unif]
Leakage-current trend	Increases with chamber and anneal temperature	TiN/HZO/TiN capacitor	–3 V; chamber 120–250 °C; anneal 400–700 °C	Not covered in sources	[park2022chamber]
Domain-switching current	Up to 80 mA	Hf _{0.5} Zr _{0.5} O ₂ capacitor	12 V; 1.1×10 ^{–3} cm ² electrode	Not covered in sources	[kim2014inductive]
Pristine P_r	0.28–31.8 μC/cm ²	TiN/HZO/TiN capacitors	120–250 °C chamber; 400–700 °C anneal	120 °C significantly degrades P_r	[park2022chamber]

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Table 3.5: Critical process and device parameter windows (*continued*)

Parameter	Range/result	Film/stack	Process condition	Limitation	Source
E _c and 2P _r	±1.5 MV/cm; about 40 $\mu\text{C}/\text{cm}^2$	TiN/HZO/ TiN 1T-1C capacitor	State-of-art FeRAM	Not covered in sources	[schroeder2023road]
E _c	0.8–2 MV/cm	Doped HfO ₂	Si, Al, Y, La dopants	Not covered in sources	[mikolajick2018mrs]
Switched polarization charge	30–60 $\mu\text{C}/\text{cm}^2$	Doped HfO ₂	Si, Al, Y, La dopants	Not covered in sources	[mikolajick2018mrs]
Residual polarization	23.94–28.18 $\mu\text{C}/\text{cm}^2$	HZO; none or dual ZrO ₂ seed	550 °C anneal	650 °C forms mon- oclinic phase	[park2023seed]
2P _r and coercive volt- age	43.4 $\mu\text{C}/\text{cm}^2$; 0.75 V	Sub-6 nm HZO/Zr-RL/ HZO	1.0 V operation	Not covered in sources	[liu2025zrrich]
Pristine and woken 2P _r	52–65 $\mu\text{C}/\text{cm}^2$	W/HZO/W; W/Pt/HZO/W	700 °C, 5 s; 15 s O ₃	Not covered in sources	[kashir2021wake]
Purge-time endurance	6×10^7 to $> 10^9$ cycles	Hf _{0.5} Zr _{0.5} O ₂ capacitors	3–90 s precursor purge	Longer purge re- duced endurance	[choi2025purge]
Temperature-controlled fatigue	Earlier onset at higher chamber temperature	TiN/HZO/TiN capacitors	120–250 °C chamber	Vacancies raise P _r but reduce cycles	[park2022chamber]
High-field breakdown cycles	SS 10 ⁵ ; SL 4×10^5	SS/SL HZO capacitors	±3.5 V cycling	Not covered in sources	[kaatranen2026unif]
Moderate-field en- durance	SS about 5×10^6 ; SL $> 10^7$	SS/SL HZO capacitors	±3.0 V cycling	SS fails earlier	[kaatranen2026unif]
Retention loss	Below 4% through 10 ⁴ s	SS/SL HZO capacitors	After 10 ⁶ cycles	SL loss slightly lower	[kaatranen2026unif]
Endurance and reten- tion	About 10 ¹¹ cycles; 82% after 10 years	Sub-6 nm HZO/Zr-RL/ HZO	1.0–1.5 V; 125 °C	Endurance is breakdown-limited	[liu2025zrrich]

3.6 Electrical Qualification and Metrology

3.6.1 Crystalline-Phase and Structural Characterization

GI-XRD and local imaging are paired because GI-XRD confirms the orthorhombic phase structurally, whereas STEM directly confirms its presence and can follow an interfacial monoclinic-to-orthorhombic transition during wake-up cycling [song2021ferro]. An orthorhombic (111) reflection at $2\theta = 30.4^\circ$ supports the diffraction assignment, but three metastable phases become indistinguishable when similar d -spacings, fine-crystallite peak broadening, and uncertain biaxial strain coincide [song2021ferro, choi2025purge]. Instrumental resolution also limits an inference from an undetected monoclinic peak; absence in the scan therefore does not establish phase exclusion [kashir2021wake]. Two-dimensional XRD supplies a processed-film stress estimate through $\sin^2 \psi$ analysis, while FTIR–ATR supplies an independent absorption spectrum; the reported sharp feature at 670 cm^{-1} is an artifact and cannot support structural interpretation [choi2025purge]. STEM additionally establishes film thickness and continuity, but HfO₂/ZrO₂ sublayers in a 5.5-nm superlattice lamella could not be distinguished when bottom-TiN roughness approached the film scale [song2021ferro, kaatranen2026unif]. Structural comparisons must therefore retain diffraction geometry, instrumental resolution, and cross-sectional specimen condition rather than equating every phase-sensitive output [choi2025purge, kim2020tin, kaatranen2026unif].

3.6.2 Film-Thickness and Uniformity Measurement

Spectroscopic ellipsometry and cross-sectional TEM provide a thickness cross-check: ellipsometry measures film and seed-layer thickness, while TEM confirms the ferroelectric-layer thickness and tests whether it changes across chamber-temperature conditions [park2022chamber, park2023seed]. Multipoint ellipsometry extends the measurement to spatial uniformity by sampling 49 wafer positions with a 3-mm edge exclusion [neuber2022peald]. Its interpretation remains reactor-specific: the standard HSO process shows an inlet-to-outlet thickness decrease in a crossflow reactor, whereas the hybrid process shows an edge-enhanced ring in a showerhead reactor [neuber2022peald]. The similar reported thickness standard deviations therefore do not imply identical spatial behavior [neuber2022peald]. In the wafer-scale SS/SL study, thickness comes from silicon reference pieces grown with the characterized wafers, so those values are not direct cross-sections of every capacitor site [kaatranen2026unif]. Comparisons consequently preserve the sampling pattern, edge exclusion, reactor geometry, and reference-piece versus device-stack distinction [neuber2022peald, kaatranen2026unif].

3.6.3 Interface and Defect Characterization

Chemical profiling and cross-sectional imaging are paired to combine composition and depth profiles from spectroscopy or SIMS with interface morphology from electron microscopy [park2022chamber, kim2020tin, liu2025zrrich, kashir2021wake]. O 1s XPS separates lattice-oxygen and sub-oxide components, while EDS supplies Hf, Zr, and O atomic fractions for chamber-temperature variants [park2022chamber]. D-SIMS provides an oxygen depth profile and confirms bottom-TiN oxidation during deposition; ToF-SIMS tracks carbon and nitrogen impurities remaining after shortened purges [kim2020tin, choi2025purge]. HRTEM then distinguishes interfacial regions at W/HZO from the sharp Pt/HZO interface, and HR-STEM supplies atomic-scale imaging of prepared lamellae [liu2025zrrich, kashir2021wake]. These outputs are complementary rather than interchangeable: laboratory XPS reaches only about 5 nm and precludes a realistic top electrode, whereas HAXPES increases depth sensitivity by one to two orders of magnitude for nondestructive vacancy profiling through electrodes [schroeder2023road]. Quantification also retains method-specific limits, because adventitious carbon and absorbed OH confound XPS-derived oxygen stoichiometry and ToF-SIMS does not quantify hydrogen ions [choi2025purge]. Interface comparisons should therefore couple morphology with chemical depth information and stop short of treating either signal alone as a complete defect description [schroeder2023road, choi2025purge, kashir2021wake].

3.6.4 Electrical Stressing and Qualification Protocols

Qualification uses a defined conditioning block followed by a specified readout, not a cycle count alone [choi2025purge, liu2025zrrich, kaatranen2026unif]. One protocol applies 5×10^3 square-wave wake-up cycles before and after measurement, while another applies 10^3 square-wave cycles before PUND; their fields or amplitudes and frequencies differ [choi2025purge, liu2025zrrich]. Endurance protocols likewise range from 3-V trapezoids with 1- μ s ramps and holds to 2.0-MV/cm bipolar pulses at 10 kHz with 2- μ s width and 20-ns edges [park2022chamber, choi2025purge]. Electrode referencing is also reversed between reported setups: one biases the TiN top electrode with the bottom grounded, whereas another biases the bottom electrode with the top grounded [kim2020tin, liu2025zrrich]. Qualification comparisons must therefore preserve conditioning state, waveform shape, amplitude or field, frequency, pulse timing, cycle count, and driven electrode [park2022chamber, choi2025purge, kim2020tin, liu2025zrrich].

3.6.5 Leakage and Breakdown Measurement

Leakage and breakdown tests separate sweep-based failure, spatial leakage mapping, and time-zero dielectric breakdown [park2023seed, kaatranen2026unif, liu2025zrrich]. The seed-layer comparison uses a 0–15-V DC sweep in 0.5-V steps, whereas wafer mapping samples leakage at ± 3.0 V against radial bottom-TiN thickness [park2023seed, kaatranen2026unif]. Excessive crystallization is associated with a leakage path and reduced breakdown in the anneal series, while temperatures above 720 °C produce large leakage and short circuits in W/HZO/W capacitors [park2023seed, kashir2021wake]. The reported TZDB comparison specifies stack and anneal variation but no waveform, electrode area, or stress time, so interpretation stops at time-zero comparison across those conditions [liu2025zrrich].

3.6.6 Hysteresis, Cycling, and Reliability Testing

Hysteresis and pulsed measurements are paired to distinguish a loop response from switched polarization [song2021ferro, schroeder2023road]. PUND is used when leakage exceeds polarization-switching current and removes leakage and dielectric contributions from the switched-polarization result, whereas P–E or P–V loops record hysteresis under their stated waveform and frequency [song2021ferro, schroeder2023road, kim2020tin, neuber2022peald]. Designed pulse trains add a switching-dynamics measurement that is distinct from loop or endurance acquisition [liu2025zrrich]. Reliability evaluation then couples cycling blocks to subsequent or interleaved readout: one seed-layer study applies rectangular cycling before triangular-pulse measurement, while the wafer-scale study uses 30- μ s up/down pulses with 5- μ s breaks and 2–3-ms triangular PUND sweeps during or after wake-up [park2023seed, kaatranen2026unif]. Wake-up and fatigue must be resolved along the cycle trajectory, because bottom- and dual-seeded devices show wake-up before fatigue after the polarization maximum, whereas unseeded and top-seeded devices fatigue without a wake-up effect [park2023seed]. Retention and high-field switching also require their full setup: the reported retention test uses 12-V, 600-ns pulses on a 1.1×10^{-3} -cm² electrode, for which severe leakage is reported, and verification below 5 nm is described as leakage dominated [kim2014inductive, mikolajick2018mrs]. Protocols differing in waveform, field, frequency, pulse width, electrode area, conditioning state, or readout method therefore support within-protocol trends, not unqualified cycle-to-cycle equivalence [park2023seed, kim2014inductive, kaatranen2026unif, liu2025zrrich].

Table 3.6: Qualification and metrology method-to-output matrix

Test/technique	Measurand/output	Specimen/stack	Stress/setup	Limitation	Source
GI-XRD	Orthorhombic peak at 30.4°	HZO thin film	Grazing-incidence diffraction	Metastable phase peaks overlap	[song2021ferro, choi2025purge]
2D XRD	Tensile stress 3.0–3.9 GPa	Processed HZO films	18° incidence	Not covered in sources	[choi2025purge]
FTIR-ATR	Absorption from 400–900 cm ^{−1}	HZO purge-series films	4 cm ^{−1} resolution	670 cm ^{−1} peak is artifact	[choi2025purge]
STEM	Orthorhombic phase and wake-up transition	Ferroelectric hafnia film	Interfacial cycling observation	Not covered in sources	[song2021ferro]
Theta–theta XRD	Ru electrode strengthens orthorhombic peak	Ni/HZO/Ru and Ni/HZO/Si	Bottom-electrode comparison	Not covered in sources	[kim2014inductive]
Spectroscopic ellipsometry	Thickness 12.64–13.37 nm	Unseeded HZO nanolaminates	Elli-SE-U measurement	Thickness varies with nanolaminate period	[park2023seed]

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Table 3.6: Qualification and metrology method-to-output matrix (*continued*)

Test/technique	Measurand/ output	Specimen/ stack	Stress/setup	Limitation	Source
Multipoint spectral ellipsometry	Mean 10.76–12.74 nm; STD 3.19–3.26%	Standard and hybrid HSO	49 points; 3 mm exclusion	Patterns depend on reactor geometry	[neuber2022peald]
Cross-sectional TEM	Thickness invariant across chamber temperatures	TiN/HZO/TiN capacitors	120–250 °C; 500 °C anneal	Not covered in sources	[park2022chamber]
O 1s XPS	Binding peaks span 532–534 eV	HZO layer	700 °C anneal	Not covered in sources	[park2022chamber]
XPS	Hf:Zr ratios 74:100–79:100	Bare HZO surfaces	3–90 s purge	Surface overlayers confound stoichiometry	[choi2025purge]
ToF-SIMS	Carbon and nitrogen impurity profiles	PEALD HZO films	Dual-beam depth analysis	Hydrogen is not quantitative	[choi2025purge]
D-SIMS	Oxygen profile; bottom-TiN oxidation	TiN/HZO/TiN stack	250 °C deposition; Cs source	Not covered in sources	[kim2020tin]
HAXPES	Subsurface vacancy profile	Electroded oxide stack	Depth gain 1–2 orders	Classical XPS reaches about 5 nm	[schroeder2023road]
HRTEM	W-interface regions; sharp Pt interface	W/HZO/W and W/Pt/HZO/W	700 °C, 5 s anneal	Not covered in sources	[kashir2021wake]
Wake-up cycling	Conditioned ferroelectric response	TiN/HZO/TiN capacitors	10 ⁵ cycles; 2.5 MV/cm	Not covered in sources	[kim2020tin]
Trapezoidal endurance cycling	Survival through 10 ⁸ cycles	TiN/HZO/TiN capacitors	3 V; 1 μs ramp/hold	Not covered in sources	[park2022chamber]
Bipolar endurance waveform	Wake-up and endurance qualification	PEALD HZO capacitors	10 kHz; 2 μs; 2.0 MV/cm	Not covered in sources	[choi2025purge]
Square-wave wake-up	Preconditioned PUND measurement	HZO; Zr-rich-layer stack	10 ³ cycles; 1.5 V; 1 kHz	Not covered in sources	[liu2025zrrich]
DC breakdown sweep	Breakdown 7.15–4.08 MV/cm	Dual-seeded HZO nanolaminates	0–15 V; 550–650 °C	Excess crystallization creates leakage paths	[park2023seed]
Radial leakage mapping	Leakage versus bottom-TiN thickness	SS- and SL-HZO wafers	±3.0 V	Electrode thickness varies radially	[kaatranen2026unif]
TZDB	Time-zero breakdown behavior	HZO; Zr-rich-layer stack	Multiple annealing temperatures	Not covered in sources	[liu2025zrrich]
Elevated-temperature I-V	Leakage reduced 20-fold	W/HZO/W and W/Pt/HZO/W	200 °C Schottky analysis	Above 720 °C causes shorts	[kashir2021wake]
PUND	Leakage-corrected switched polarization	Thin ferroelectric films	Positive-up negative-down pulse sequence	Depolarization fields cause underestimation	[song2021ferro, schroeder2023road]
P-E hysteresis	2P _r and E _c	TiN/HZO/TiN capacitors	10 kHz	Not covered in sources	[kim2020tin]
P-V hysteresis	Polarization before and after wake-up	HZO; Zr-rich-layer stack	1.75 V operation	Not covered in sources	[liu2025zrrich]
Fatigue cycling	P _r falls to 5 μC/cm ²	Capped 25 nm HZO	1 MHz; 8 V; 1000 ns	Inferior to SBT after 10 ⁹ cycles	[kim2014inductive]

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Table 3.6: Qualification and metrology method-to-output matrix (*continued*)

Test/technique	Measurand/ output	Specimen/ stack	Stress/setup	Limitation	Source
Seed endurance cycling	Stable through 10 ⁷ –10 ⁸ cycles	Seeded HZO nanolaminates	550 °C; rectangular pulses	Fatigue follows 10 ⁵ cycles	[park2023seed]
Wafer-scale PUND	2P _r spans 27.6–31.5 $\mu\text{C}/\text{cm}^2$	SS- and SL-HZO capacitors	3.0 V; 150 mm wafer	RSD spans 2.2–7.0%	[kaatranen2026unif]
Wake-up cycling	Wake-up-free 2P _r near 64 $\mu\text{C}/\text{cm}^2$	W/HZO/W capacitor	3 MV/cm; 100 Hz; 1000 cycles	Short ozone pulses pinch loops	[kashir2021wake]
Pulse-switching measurement	Switching dynamics	HZO/Zr-rich-layer/HZO	Designed pulse-train sequence	Not covered in sources	[liu2025zrrich]

3.7 Ontology Model and Communication

3.7.1 Protégé Ontology Design

3.7.2 Classes and Properties

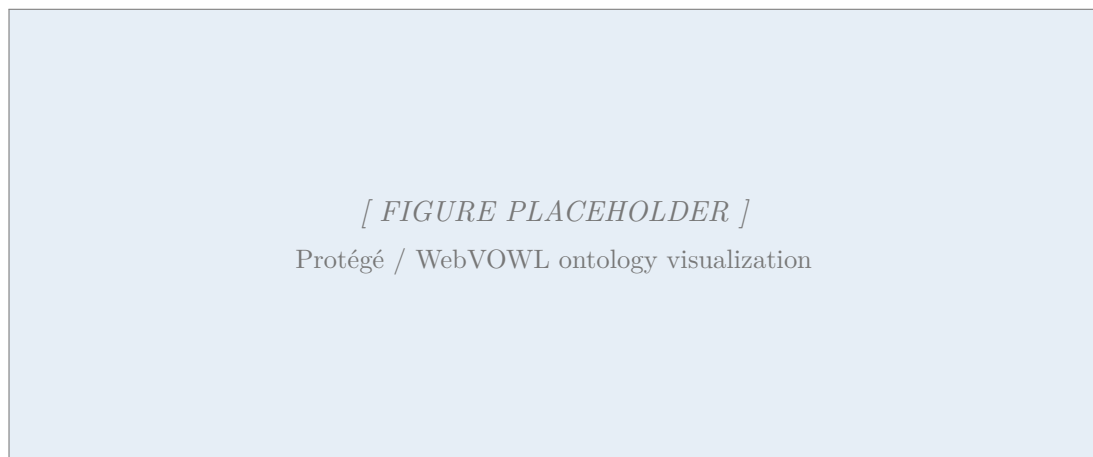
Table 3.7: ALD HfO₂ ontology classes and properties

Ontology element	Element type	Domain /parent	Range /definition

3.7.3 Individuals and Example Records

Table 3.8: ALD HfO₂ ontology individuals

Individual	Class	Key assertions	Source record

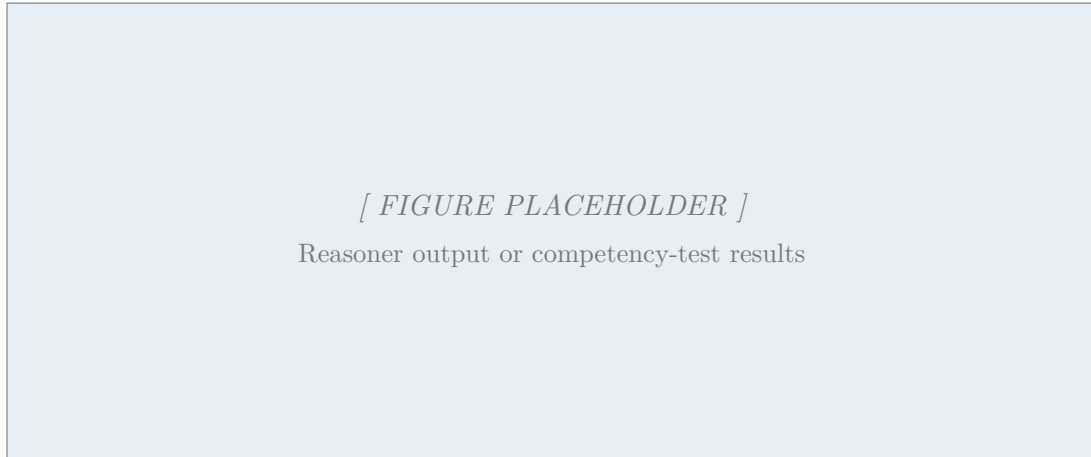
Figure 3.2: ALD HfO₂ ontology visualization

3.7.4 Digital Reference Mapping

Table 3.9: ALD HfO₂ Digital Reference mappings

Local concept	Digital Reference IRI	Mapping relation	Evidence and rationale
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3.7.5 Reasoner and Validation Results

Figure 3.3: ALD HfO₂ ontology validation results

3.7.6 Ontology Limitations

Chapter 4

Backend / BEOL Integration

4.1 Module Purpose

4.2 Input Condition

4.3 Step-by-Step Flow

4.3.1 Pre-Metal Dielectric and Planarization

4.3.2 Contact Formation

4.3.3 Metal 1 Formation

4.3.4 Via Formation

4.3.5 Repeated Interconnect Stack

4.3.6 Upper-Level Global Routing

4.3.7 Passivation and Pad Opening

4.3.8 Wafer Probe and Test Handoff

Table 4.1: Ordered Backend / BEOL process steps

Order	Process step	Purpose	Method	Equipment type	Output /next input
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4.4 Key Materials and Equipment

Table 4.2: Backend / BEOL materials and equipment

Step	Material or chemical	Equipment or capability	Role in the module
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4.5 Critical Parameters

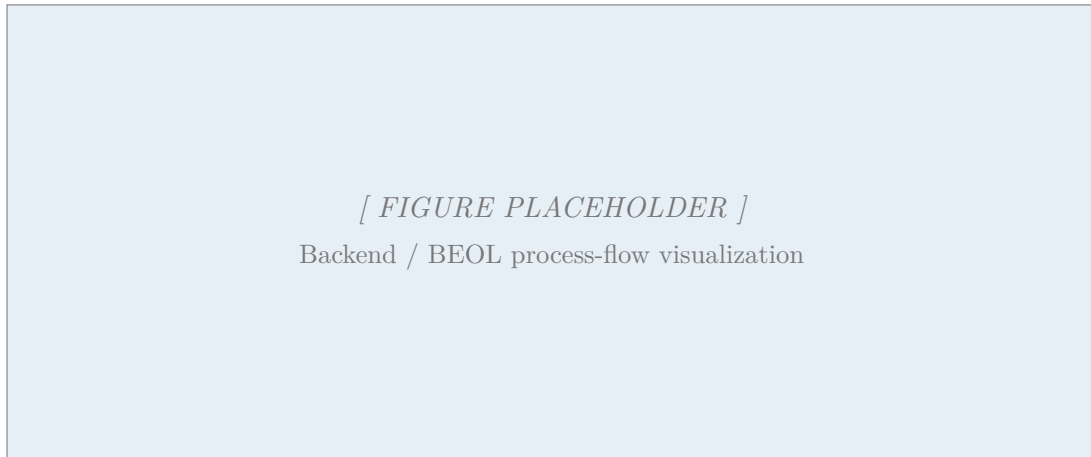


Figure 4.1: Backend / BEOL module process flow

4.5.1 Contact and Via Resistance

4.5.2 Line Resistance and Capacitance

4.5.3 Overlay and Critical Dimensions

4.5.4 CMP Planarity

4.5.5 Thermal Budget

Table 4.3: Backend / BEOL critical parameters

Step	Parameter	Unit	Value or range	Evidence status /source
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4.6 Metrology and Electrical Tests

4.6.1 Contact and Via-Chain Tests

4.6.2 Line-Resistance and Defect Inspection

4.6.3 CMP Thickness and Planarity Mapping

4.6.4 RC and Signal-Integrity Tests

4.6.5 Electromigration and Dielectric Reliability

4.6.6 Final Wafer Test

Table 4.4: Backend / BEOL metrology and electrical tests

Step or checkpoint	Measurement or test	Acceptance measure	Decision or process response
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4.7 Ontology Model and Communication

4.7.1 Protégé Ontology Design

4.7.2 Classes and Properties

Table 4.5: Backend / BEOL ontology classes and properties

Ontology element	Element type	Domain /parent	Range /definition
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4.7.3 Individuals and Example Records

Table 4.6: Backend / BEOL ontology individuals

Individual	Class	Key assertions	Source record
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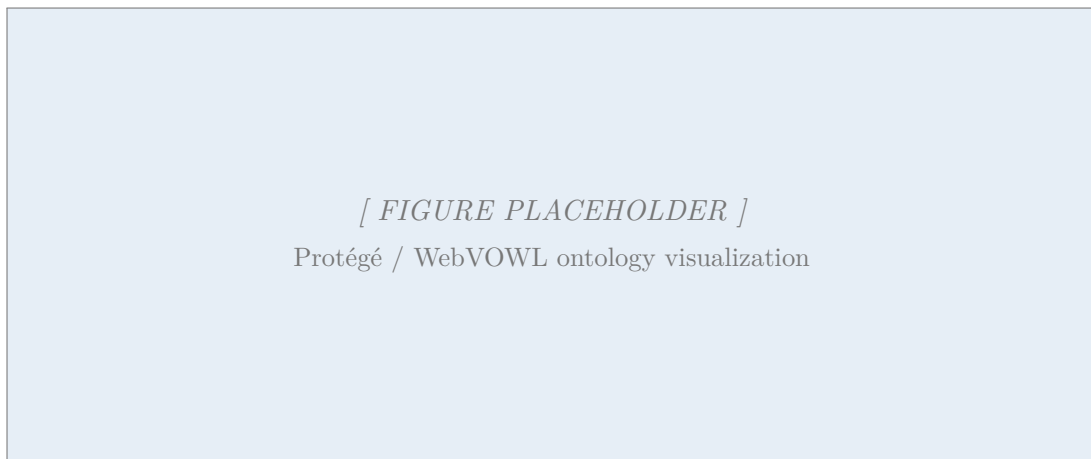


Figure 4.2: Backend / BEOL ontology visualization

4.7.4 Digital Reference Mapping

Table 4.7: Backend / BEOL Digital Reference mappings

Local concept	Digital Reference IRI	Mapping relation	Evidence and rationale
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4.7.5 Reasoner and Validation Results

4.7.6 Ontology Limitations

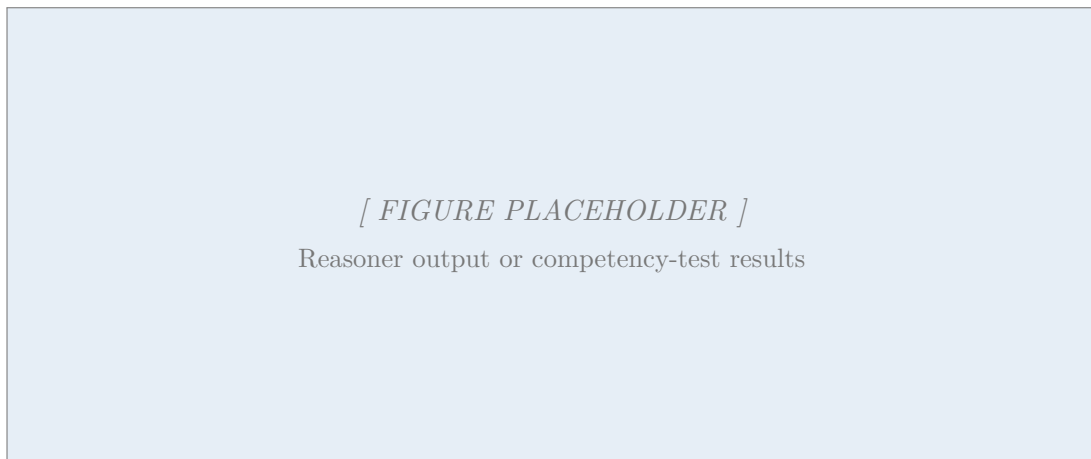


Figure 4.3: Backend / BEOL ontology validation results

Chapter 5

Ontology and Digital Reference Method

5.1 Ontology Engineering Method

5.2 Digital Reference Alignment Method

Chapter 6

Evaluation and Conclusion

6.1 Process and Ontology Coverage

6.2 Limitations